



Directional Drilling Services Workshop

Mud Motor and Drilling Jar

Halliburton Drilling Systems Indonesia

June 2026

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Mud Motor

└ Positive Displacement Mud Motors (PDM)



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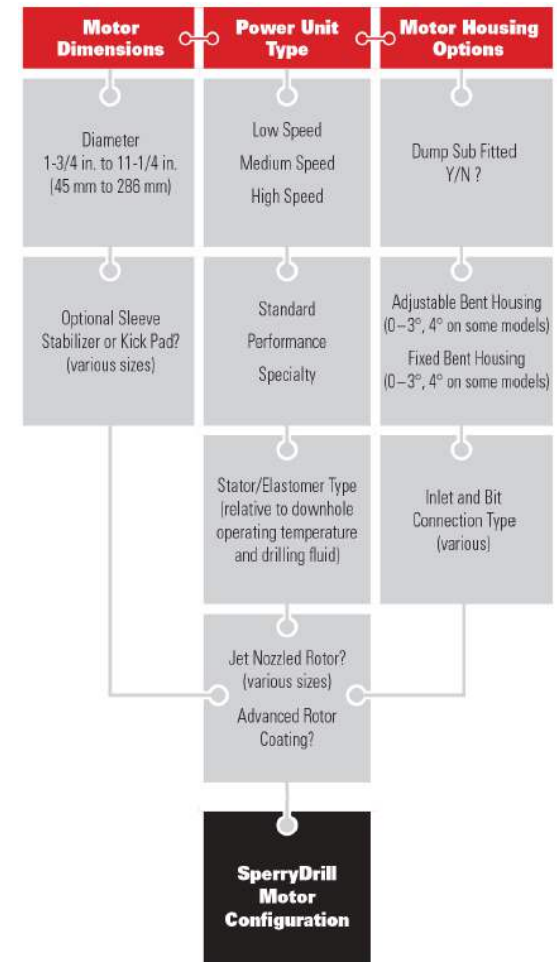
Positive Displacement Mud Motors

What are the Sperry Drill motors?

- Most reliable mud motors in the market
- Proprietary design: increased efficiency, added reliability and simplified maintenance processes
- Offered in a wide variety of rotor/stator/stages configurations
- Customized Halliburton bits match motor specifics and provide the best drilling performance

Speed Ranges

High Speed	195		1375 rpm
Medium Speed	60		480 rpm
Low Speed	30		360 rpm

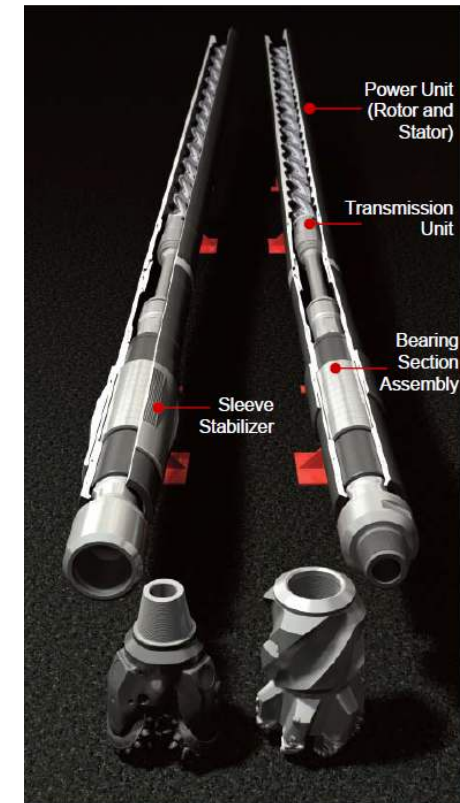


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What Positive Displacement Mud Motor is and how it works

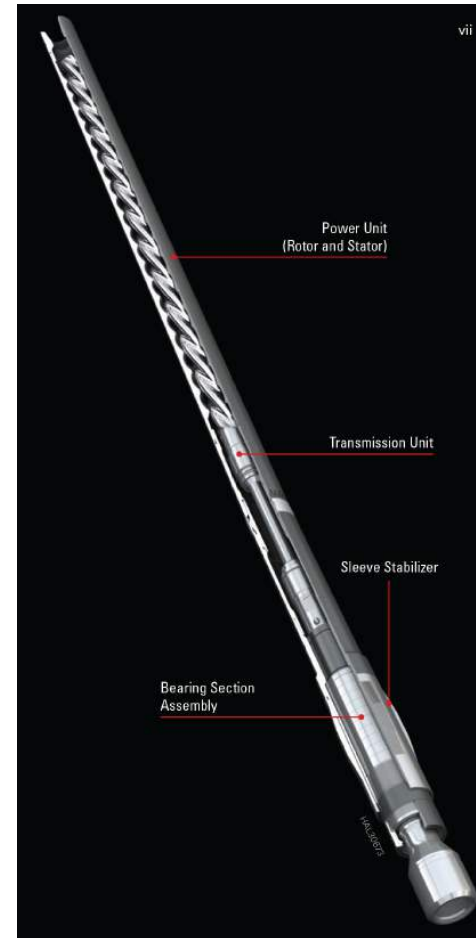


- The mud motor is a progressive cavity positive displacement pump.
- The motion of the rotor in the stator produces discrete cavities between the rotor and stator.
- The power section of a positive displacement motor (PDM) **converts the hydraulic energy of high-pressure drilling fluid to mechanical energy** in the form of torque output and RPM for the drill bit
- Additionally, to the power section, the PDM contains a transmission unit, Bearing Pack section assembly and a Bend section which are designed to provide optimum performance in all drilling application types.



Mud Motor Components

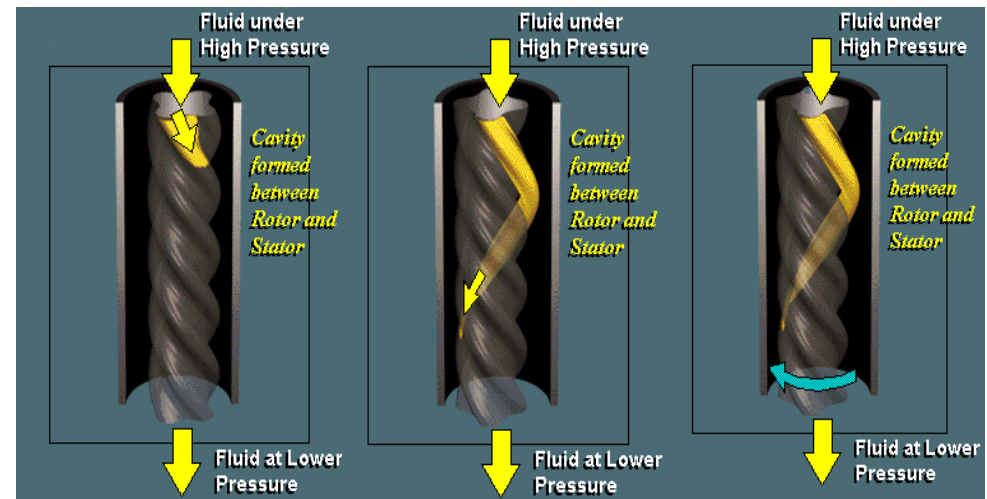
- **Power Unit**
 - Rotor
 - Stator
- **Transmission Unit**
 - Bend Angle
- **Bearing Section Assembly**
 - Flow Restrictor
 - Sleeve Stabilizer



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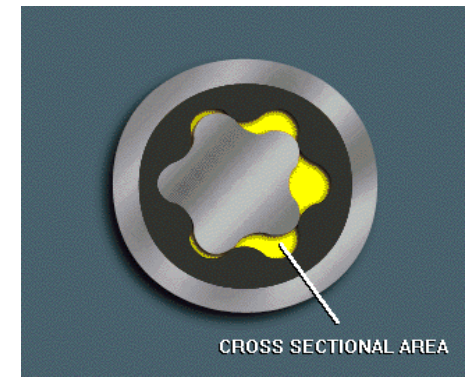
Motor Power Unit (Rotor/Stator)

- During drilling operations, **high-pressure fluid is pumped into the top end of the power section**, where it fills the first set of open cavities.
- The differential pressure across two adjacent cavities forces the rotor to turn, and as this occurs, adjacent cavities are opened allowing the fluid to flow progressively down the length of the power section.



Motor Power Unit (Rotor/Stator)

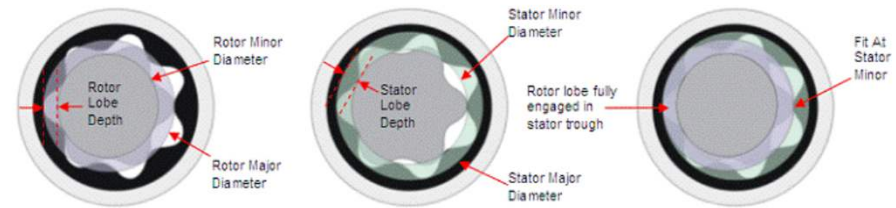
- The differential pressure causes drilling fluid to enter the cavities at the top of the motor. As it moves through the motor, the fluid pushes on the rotor causing it to rotate.
- This action converts the fluids hydraulic energy into mechanical energy (torque and RPM) which is transferred through the remainder of the motor to the drill bit.



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Motor Power Unit (Rotor/Stator)

- Upon assembly, the rotor and stator 'mate' and produce helical fluid cavities.
- Correct compression of the elastomer by the rotor ensures good seal lines are created between the rotor lobe and stator lobe.
- If the interference fit is too loose then mud is allowed to bypass between the two lobes. As this occurs the motor is not able to produce the expected torque and RPM required to drill.

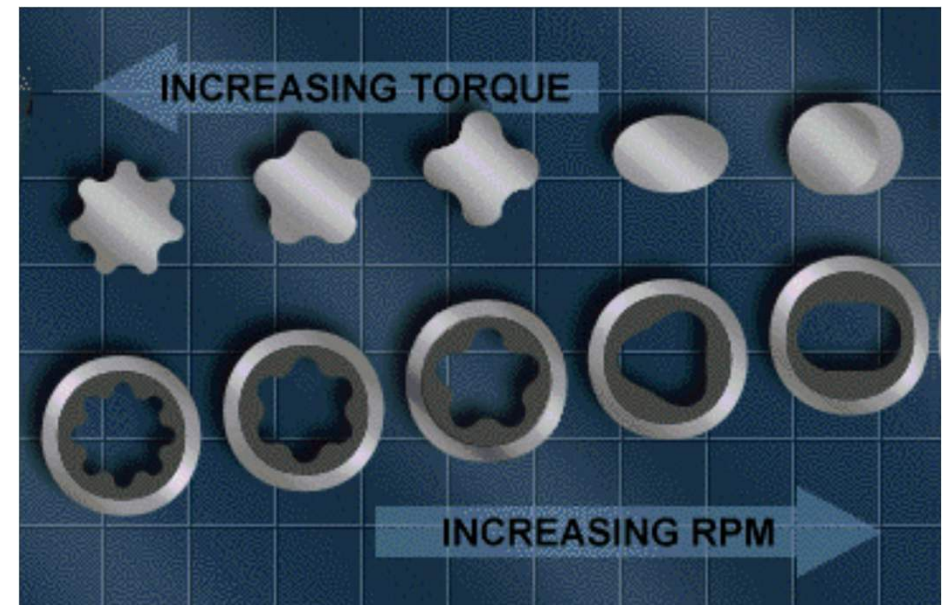


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Motor Power Unit (Rotor/Stator)

Lobe Configuration

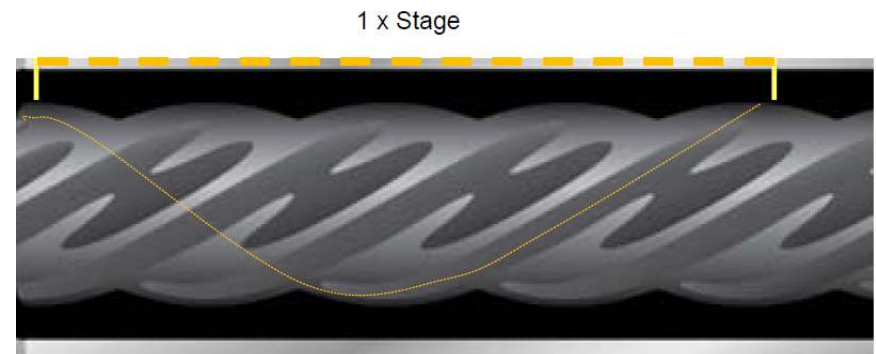
- The speed and torque of a power section is directly linked to the number of lobes on the rotor and stator.
- So the higher the number of lobes, the higher the torque and the lower the RPM.
- The lower the number of lobes, the lower the torque, and the higher the RPM.



Motor Power Unit (Rotor/Stator)

Rotor Stage Configuration

- A stage in a power section is defined as the distance, measured parallel to the stator axis, between two corresponding points of spiral lobes.
- The number of stages affects, directly proportional, the torque output of the motor.
- The more stages the PDM contains, the higher the torque output is if they are the same Motor size.



Motor Power Unit (Rotor/Stator)

Factors the power section Torque and RPM

- 1. Configuration** – The number of Lobe, stage and others will determine the displacement characteristic of that design
- 2. Flow Rate** – It has a direct effect. As the flow rate increases, the RPM will increase.
- 3. Differential Pressure** – Increasing the differential pressure will increase slippage and eventually decrease RPM output.
- 4. Temperature** - High downhole temperatures affect the fit between the rotor and the elastomer (rubber) stator, resulting in increased loading of the elastomer.

Transmission Unit C.V. (Constant Velocity) Assembly

- The C.V. is the link between the rotor and the bearing assembly.
- Converts the eccentric motion of the rotor into the concentric rotary motion for the driveshaft.
- Transmits the torque and the rotary motion of the rotor, created by the power section, to the bearing assembly.
- The hydraulic down thrust of the rotor is also transferred to the bearing section through this shaft.



Transmission Unit C.V. (Constant Velocity) Assembly Adjustable Housing Assembly



- Works on an offset pin and box concept
- Typically adjust from 0 to 3 degrees

Adjustable Housing Bend Increments (in degrees)							
Up to 9-5/8 in. Motor (not 5 in. and 7 in.)	0.39	0.78	1.15	1.50	1.83	2.12	
	2.38	2.60	2.77	2.89	2.97	3.00	
Legacy Sperry Design 5 in. and 7 in. Motors	0	0.31	0.62	0.93	1.22	1.50	2.00
	2.23	2.43	2.60	2.74	2.85	2.93	3.00
11-1/4 in. Motor	0.26	0.52	0.77	1.00	1.22	1.41	
	1.59	1.73	1.85	1.93	1.98	2.00	



Bearing Section Assembly



- The bearing assembly consists of:
 - a thrust bearing assembly
 - radial bearings
 - drive shaft.
- Additionally, there is a flow restriction between the bearing/ motor housing and sleeve stabilizer.
- The thrust bearings deal with down & upward (axial) loading while the Tungsten Carbide coated radial bearings either side of the thrust bearings absorb lateral side loading from the driveshaft.
- *Ball bearing aren't reuseable*



Bearing Section Assembly

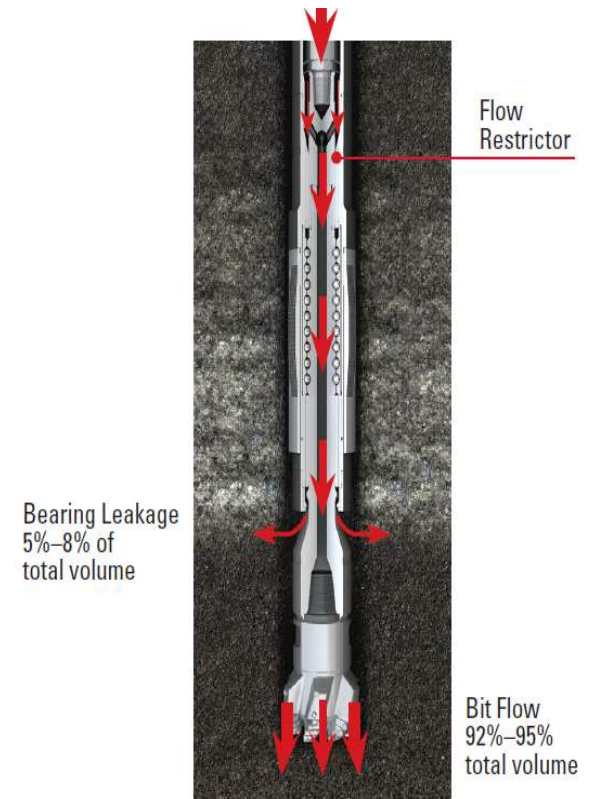


- The thrust bearings are designed to sustain the applied weight to the drill bit while on bottom (Upward Axial). They are also capable of bearing the downward hydraulic thrust load of the rotor (Downward Axial).
- The radial bearings are employed above and below the thrust bearings to absorb lateral side loading of the drive shaft.
- The drive shaft transmits both axial and torsional loading to the bit. The bit is connected to the drive shaft by the bit box.
- The correct balance between flow rate and the reactive force of WOB applied will prolong the operation motor life



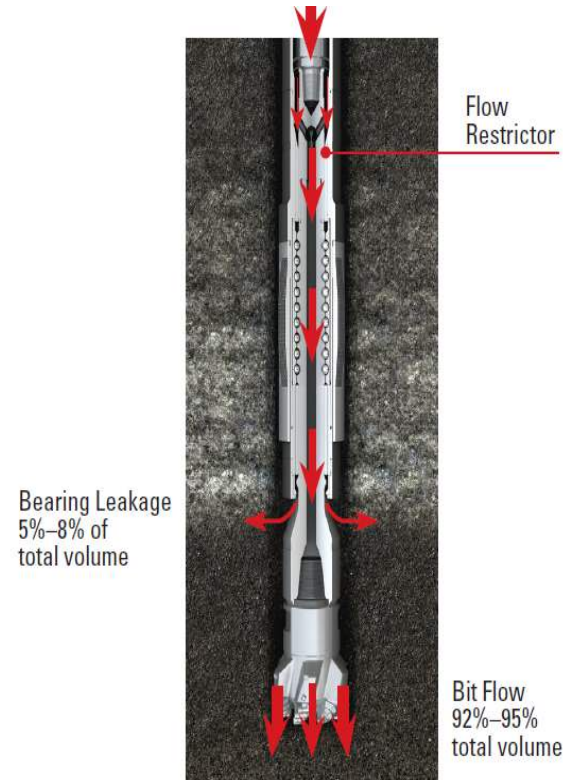
Bearing Section Assembly

- The upper stationary radial bearing also acts as a **flow restrictor** allowing the correct amount of drilling fluid to enter the bearing pack.
- The Lower rotating radial bearing also acts as a flow restrictor ensuring that the drilling fluid does not flow through the bearing pack too quickly.
- The drilling fluid-lubricated bearing assemblies operate on approximately 5 to 8% of the circulating fluid. However, certain motor can have up to 12 – 15% bypass.



Bearing Section Assembly

- Minimum 200psi bit pressure drop for proper mud motor lubrication (approximately 5 to 8% of the circulating fluid. certain motor can have up to 12 – 15% bypass.)
- Fluid Level drop isn't a signal of mud motor malfunction



Bearing Section Assembly

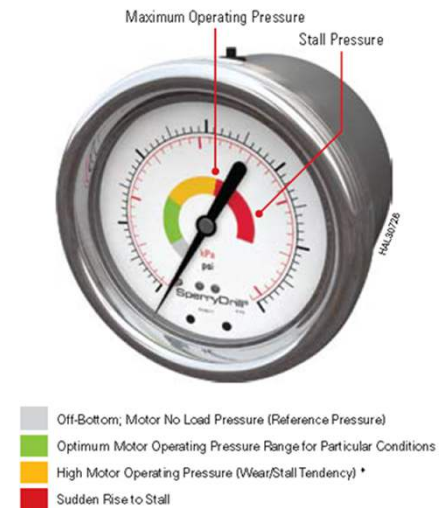


- The bearing housing could come to a Sleeve Stabilizer or, just a protected sleeve.
- It gauge of the stabilizer will depend the DLS required for the project.

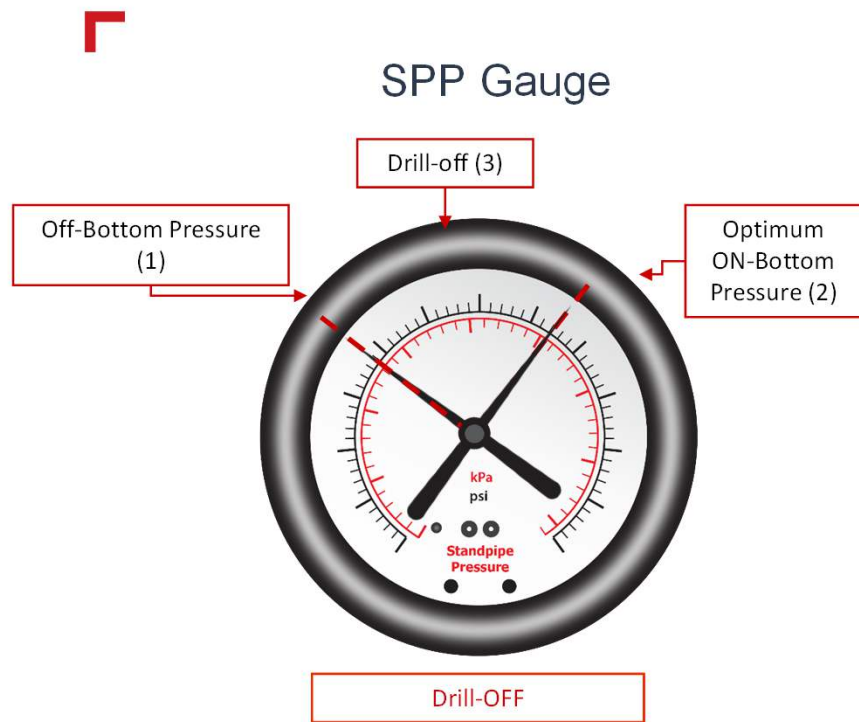


Motor Operation Differential Pressure

- Operating Differential Pressure is the pressure drop that occurs across the rotor and stator as torque is produced to drive the bit.
- The Differential Pressure produced across the rotor and stator is directly proportional to the torque applied at the bit and the reactive torque that acts upon the stator elastomer (100% differential pressure equates to 100% torque output).
- Increasing WOB applied to a motor normally results in an increased motor operating differential pressure and increased output torque.

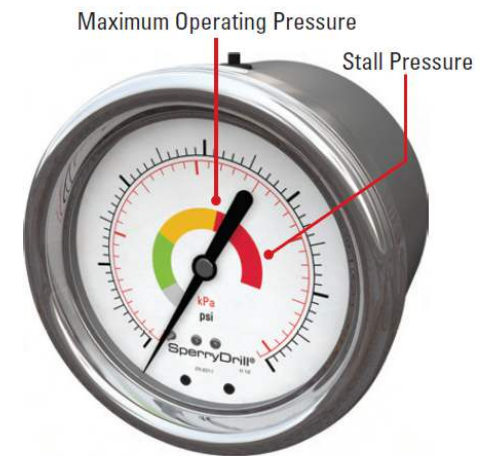
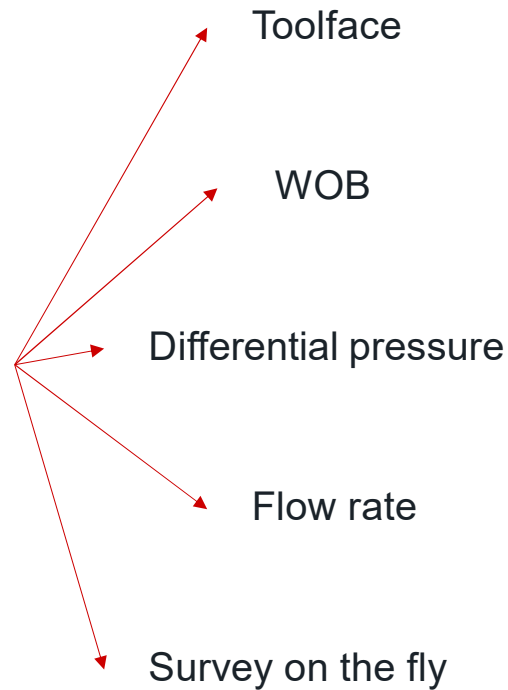


Motor Operation Differential Pressure



- When a motor is run off-bottom at 'No Load' or 'Free-Running'. Off-Bottom pressure represents the No-Load + BHA+ DP drop pressure (1).
- As it is lowered to the hole bottom, the bit contacts the formation, weight is applied to the bit and the Standpipe pressure will rise due to the increased torque demand as the bit begins to drill (2).
- As formation penetration occurs, the weight on bit and differential pressure (and torque) will reduce or 'drill off' (3).
- Therefore, keep a constant Differential Pressure, for an acceptable ROP, will improve directional performance.

Monitor Drilling Parameter



- Off-Bottom; Motor No Load Pressure (Reference Pressure)
- Optimum Motor Operating Pressure Range for Particular Conditions
- High Motor Operating Pressure (Wear/Stall Tendency) *
- Sudden Rise to Stall

* The High Motor Operating Pressure Zone shown occurs at high loadings where drilling conditions are less than optimum. Within this zone, wear rate and stall tendency is increased (See 2.2.1 and 2.2.2).



Extended Learning

DOWNHOLE MOTOR

• • • •

Sliding

Typical Terraforce Motor

Motor Size	Top Connection	Bottom Connection	Lobe/Stage	Flow Range (GPM)	RPG
TerraForce Motor 9-5/8"	6 5/8 REG BOX	7 5/8 REG BOX	6/7 ML 5.0 STG	600 - 1200	0.13
TerraForce Motor 8-00"	6 5/8 REG BOX	6 5/8 REG BOX	6/7 ML 4.0 STG	300 - 900	0.17
TerraForce Motor 7-00"	4 1/2 IF BOX (NC 50)	4 1/2 REG BOX	7/8 ML 6.0 STG	300 - 675*	0.31
TerraForce Motor 6-3/4"	4 1/2 IF BOX (NC 50)	4 1/2 REG BOX	6/7 ML 5.0 STG	300 - 600	0.29
TerraForce Motor 5.00"	3 1/2 IF BOX (NC 38)	3 1/2 REG BOX	6/7 ML 6.0 STG	150 - 350	0.80
TerraForce Motor 4-3/4"	3 1/2 IF BOX (NC 38)	3 1/2 REG BOX	7/8 ML 3.8 STG	150 - 250	0.56



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Drilling Ops : Best Practice

Back-Reaming operation Recommendation

- Reduce RPM 50-70% of drilling RPM
- Reduce Flowrate 30% of Maximum Motor Specification - Recommended

TerraForce RPM limitation vs bend setting

- Reduction on the maximum RPM for 1.50° bend and avoid any rotation for 1.83° TerraForce motor.

Motor size (inches)	Motor BH (°)	Curve On bottom / Reaming (RPM)	Tangent or Vertical On Bottom / Reaming (RPM)	Circulate hole clean after TD In a Curve section / straight section (RPM)
9 5/8	<1.15	80 / 60	90 / 60	60 / 80
	1.5	30 / 30	60 / 40	40 / 60
	1.83	No Rotation	No rotation (check <u>MaxBHA</u>)	10 / 20 (Preferable, no rotation, risk of fatigue damage)
8	<1.15	80 / 60	90 / 60	60 / 90
	1.5	40 / 40	60 / 60	40 / 60
	1.83	No rotation	No rotation (check <u>MaxBHA</u>)	10 / 20 (Preferable, no rotation, risk of fatigue damage)
6 3/4 - 7	<1.15	80 / 60	90 / 60	60 / 90
	1.5	40 / 40	60 / 60	60 / 70
	1.83	No rotation	No rotation (check <u>MaxBHA</u>)	10 / 20 (Preferable, no rotation, risk of fatigue damage)
4 3/4	<1.15	80 / 60	90 / 60	60 / 90
	1.5	40 / 40	60 / 60	60 / 80
	1.83	40 / 20 (<5 DLS)	40 / 20 (check <u>MaxBHA</u>)	20 / 40 (check <u>MaxBHA</u>)

Notes

- Above table is a recommendation, but final confirmation shall come from the results of BHA modeling for Bending Moments (please refer to the well's DoS, Engineering section for Bending moment results and recommendations).
 - Motor handbook can be used as a first reference. Motor should be operated in the "Green Zone". Operating in "Ambar Zone" will apply sufficient stress in the stator connections and other components to initiate fatigue damage. Once induced, fatigue propagates at a rate determined by the number of subsequent stress cycles (RPM) and the level of stress in each cycle.
 - Once fatigue is initiated can only be repaired by machining (cutback) of the connection or scrapping the component if it is not repairable
- 1.83 BH, avoid rotation. For final hole cleaning, if well is < 35 deg, the circulation can be done with reciprocation and without rotation. For >35deg apply low RPM while reciprocating pipe.
- For BH > than 1.83 will be managed in a case-by-case basis. This type of BH shall not be rotated.
- Any deviation must be approved via MOC prior client's written agreement on covering equipment repair or replacement base on the extend of the damage.
- In case MaxBHA shows bending moments are not exceeded in any part, the drillstring rotation can be applied up to 80 RPM for drilling operations (Given favorable conditions, RPM can be increased to 100).
- For Backreaming operations it is recommended to reduce RPM by 50%.

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Drilling Operation: Lesson Learn

LCM Plugging Prevention

- Limited on MWD System 40 lb/bbl for positive pulse Telemetry

Recommendation :

- LCM displaced by fresh mud with **Minimum 1.5 string volume** to avoid any left LCM inside the BHA.
- It is suggested once displacement finished to POOH 2-3 stand then perform flushing or circulating while waiting LCM settling to ensure no LCM left inside string.
- Released IOM – Pump LCM thru BHA Operation Procedure



Drilling Operation: Lesson Learn

Fluid Level drop on Mud Motor and float valve

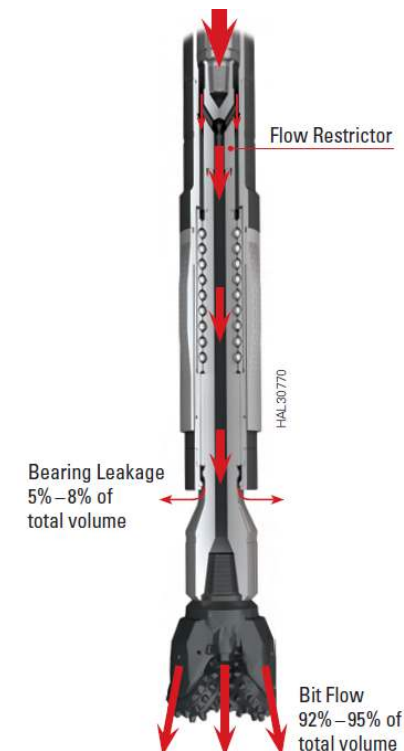
- 5 to 8% of the circulating fluid bypass to bearing assembly. Certain motor can have up to 12 – 15%.
- Float valve is to prevent fluid from annulus (back flow) to string
- Fluid Level drop is not a sign of float sub and mud motor malfunction

Motor change out criteria based on :

- Hours
- Bearing gap
- ROP - Differential pressure when drilling
- Or any certain condition, such as plugged or chunking.



Motor O.D.	New Motor		Previously Run Motor	
	in.	mm	in.	mm
4-3/4	0.063	1.6	0.125	3.2
5	0.063	1.6	0.125	3.2
6-1/4	0.081	2.1	0.162	4.2
6-1/2	0.088	2.2	0.175	4.4
6-3/4	0.088	2.2	0.175	4.4
7	0.088	2.2	0.175	4.4
8	0.113	2.9	0.230	5.8
9-5/8	0.125	3.2	0.250	6.4
11-1/4	0.140	3.6	0.310	7.9



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Directional Drilling Services Workshop
Drilling Jar

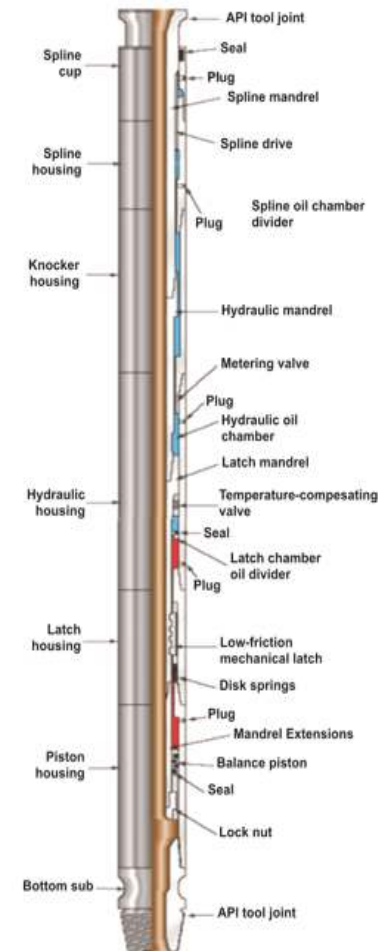
Halliburton Drilling Systems Indonesia

Nov 2024

Drilling Jar

Functionality

- Impact tool located in the drill string used to free stuck pipe
- Most effective when the drill string is mechanically restrained by the formation and not differentially stuck due to mud imbalance
- Types of drilling jars; each refers to trigger mechanisms available
 - Hydraulic
 - Mechanical
 - HydroMechanical



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Drilling Jar

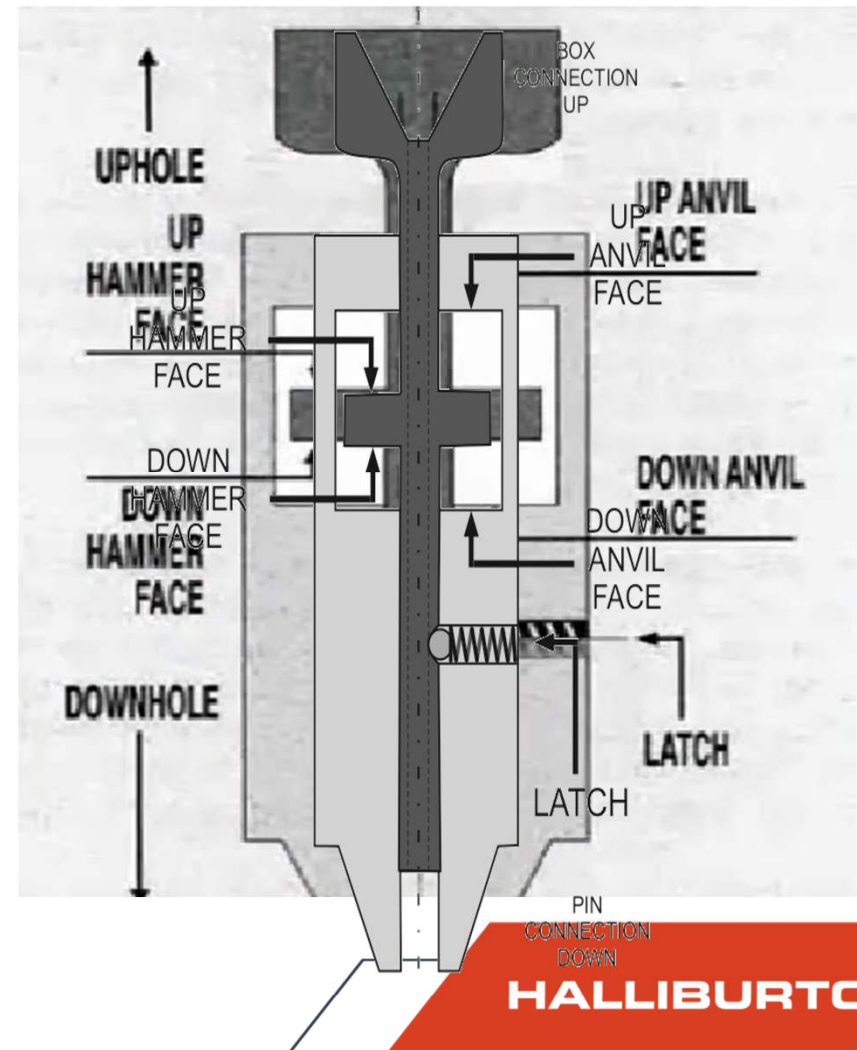
Mechanical Jar

- Trigger is mechanical
- The load to trip the trigger up or down is preset
- Will only trip when the applied load exceeds the trigger setting
- Fires immediately
- Is normally used latched at mid stroke ready to jar up or down

• Application

- Vertical Wells
- Directional Wells with Less Than 30° Hole Angle
- Geothermal Wells

PRINCIPLE OF MECHANICAL JAR



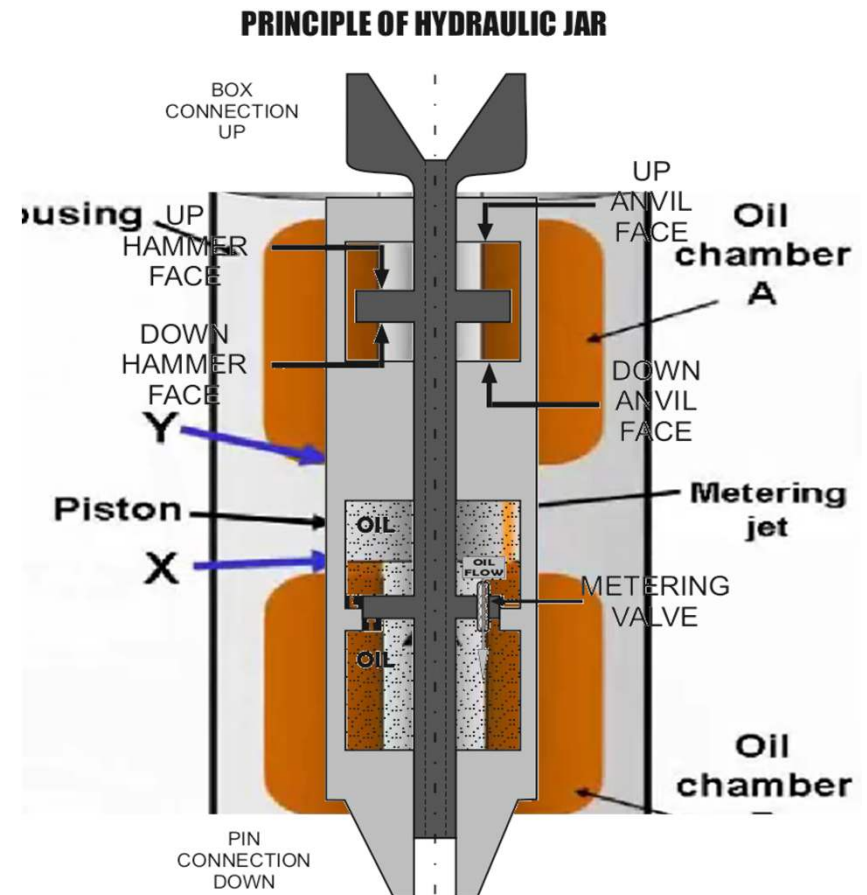
Drilling Jar

Hydraulic Jar:

- Has the same free stroke / anvil and hammer
- Between the up and down stroke is a metering (delay) valve
- When the load is applied to the jar, it moves a piston in a cylinder. This forces (meters) oil slowly from one side of the piston to the other through the metering valve
- At the end of the metering stroke the piston releases the hammer which strikes the anvil and generates the impact

Application

- Vertical
- Directional wells with elevated torque and drag
- Horizontal/extended reach wells



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Drilling Jar

Hydro-Mechanical Jar (Sperry Sledgehammer Jar):

- Hydraulic up – Mechanical down jar
- Has a mechanical latch at mid-stroke which keeps the jar ready for immediate up or down jarring and eliminates risk of accidental jarring
- This jar has an important extra feature, which is an additional mechanical latch at mid-stroke. This keeps the jar ready for immediate up or down jarring and eliminates the risk of accidental jarring during all normal drilling operations.
- Sledgehammers can be shipped with a 'latch ratio' of 2:1 (uplatch setting is twice the downlatch) or with a ratio of 3:1.

Drilling Jar

• Jar Placement

- For all applications (vertical, deviated and horizontal), the jar should be placed as close as possible to the predicted stuck point on the BHA (80% of stuck pipe incidents is across the BHA)
- From a jarring dynamics point of view, a jar high up the string (inside the cased hole) is not as effective for freeing a stuck BHA as a jar in/on the top of the BHA
- **Ensure Jar is not place in neutral point** Ideally WOB is applied from weight below Jar, but we can apply more WOB than weight below jar, means jar in compression while drilling
- Has enough weight to do Jar down, without creating buckling

Drilling Jar

• Operations Guide - To Jar Down

- Set down enough string weight on the jar to unlatch. It will fire immediately.
- To jar again, recock by picking up until the jar begins to take an overpull (greater than 5000lbs (2250daN) but less than the uplatch setting), then repeat.
- When the jar has been fired and before it is recocked, the jar free stroke can be used to establish an accurate measurement of the frictional drag acting on the free portion of the drill string.

Drilling Jar

• Operations Guide - To Jar Up

- Pick up enough string tension (overpull) to unlatch and start the hydraulic delay, adjust overpull to the desired level, and wait for the jar to fire. This will take some 1040 seconds ***depending*** on the tension set and downhole temperature.
- To jar again, first recock by slacking off until the jar begins to take weight (more than 5000lbs (2250daN) but less than the downlatch value) then repeat. The unlatching and re-latching will be seen as a flutter on the Martin Decker hook load.

Typical Drilling Jar



Motor Size	Top Connection	Bottom Connection	RATIO
Sledgehammer Jar 8"	6 5/8 REG BOX	6 5/8 REG PIN	RATIO 2:1
Sledgehammer Jar 8"	6 5/8 REG BOX	6 5/8 REG PIN	RATIO 3:1
Sledgehammer Jar 6 3/4"	4 1/2 IF PIN (NC 50)	4 1/2 IF BOX (NC 50)	RATIO 2:1
Sledgehammer Jar 6 3/4"	4 1/2 IF PIN (NC 50)	4 1/2 IF BOX (NC 50)	RATIO 3:1
Sledgehammer Jar 6 1/2"	4 1/2 IF BOX (NC 50)	4 1/2 IF PIN (NC 50)	RATIO 2:1
Sledgehammer Jar 6 1/2"	4 1/2 IF BOX (NC 50)	4 1/2 IF PIN (NC 50)	RATIO 3:1
Sledgehammer Jar 4 3/4"	3 1/2 IF BOX (NC 38)	3 1/2 IF PIN (NC 38)	RATIO 2:1

RATIO 2:1 Up : 80.000LBS

RATIO 3:1 Up : 120.000LBS

Down: 40.000LBS

Down: 40.000LBS

DELAY 50 SEC

DELAY 50 SEC

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Typical Drilling Jar

Sample Cases

ACS-24 BNK PX17B_LQR (N2-W3-73B) 8.5in BHA

			
BHA Weight	(klb)		(klb)
in Air (Total)	: 48.45	in Mud (Total)	: 41.65
in Air (Below Jars)	: 20.71	in Mud (Below Jars)	: 17.81
		Mud Weight (ppg)	: 9.20

DAILY PROGRESS		Date : 11-Nov-24			Well Name		BNK_PX17B_LQR (N2-W3-73B)		Depth In (ftMD)			470 ftMD		LAS					
	Dist	%	Time(hr)	ROP(ft/hr)	Field Name	Bangko			Depth Out (ftMD)	1929 ftMD		Svg Depth	1929.00						
SLIDE	0	0.0%	0.00 Hrs		RIG	ACS-24			Drill Hrs	9.88 Hrs		Inclination	20.40 deg						
ROTATE	536	100.0%	5.18 Hrs	103.41	BHA Type	7" Mud Motor_BH: 0.78 deg			Total Slide(ft)	0 (0.0%)		Azimuth	257.40 deg						
Tot. Drilled	536	ROP AVG	103.41 ft/hr		Hole Size	8-1/2" (TFA : 0.982 in2)			Total Drilled (ft)	1459 ft		TYD	1832.10						
Date	Time		Depth		Drilling Mode	Toolface	Slide Dist.	Rotate Drilled	ROP ft/hr	Drill Time (Hrs)	RPM	WOB (Klbs)	Flow Rate	Off Btm Press	On Btm Press	P/U Weight	S/O Weight	Rot. Weight	Drilling Torque Off/On(Kft-lbs)
11-Nov-24	From	To	From	To															
	11:49	11:54	1888	1893	Rotary			5	60.0	0.08	80	4-6	550	1300	1400	86	71	78	2.1-2.9
	13:00	13:04	1893	1898	Rotary			5	75.0	0.07	80	4-6	550	1300	1400				
	13:21	13:26	1898	1903	Rotary			5	60.0	0.08	80	4-6	550	1300	1400				
	13:39	13:41	1903	1906	Rotary			3	90.0	0.03	80	4-6	550	1300	1400				
	13:54	13:56	1906	1909	Rotary			3	90.0	0.03	80	4-6	550	1300	1400				
	14:02	14:04	1909	1912	Rotary			3	90.0	0.03	80	4-6	550	1300	1400				
	14:05	14:07	1912	1915	Rotary			3	90.0	0.03	80	4-6	550	1300	1400				
	14:12	14:15	1915	1918	Rotary			3	60.0	0.05	80	4-6	550	1300	1400				
	15:42	16:20	1918	1929	Rotary			11	17.4	0.63	80	4-6	550	1300	1400	87	71	79	2.2-3.0

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Typical Drilling Jar

Sample Cases

ACS-26 P_PETA24-P04 (N1-W1-35B) 8.5in BHA

HALLIBURTON Sperry Drilling	
BHA Weight (klb)	(klb)
in Air (Total) : 81.94	in Mud (Total) : 71.07
in Air (Below Jars) : 20.88	in Mud (Below Jars) : 18.11
Mud Weight (ppg) : 8.70	

Date :	10-Nov-24	Well Name	: PETA24-P04 (N1-W1-35B)		Depth In (mMD)	3261.00 ft		
Dist	%	Time(hr)	ROP(ft/hr)	Field Name	Depth Out (mMD)	5483.60 ft	Svg Depth	5483.60
0	0.0%	0.00 Hrs		RIG	Drill Hrs	66.28 Hrs	Inclination	23.66 deg
300	100.0%	17.48 Hrs	17.14	BHA Type	Total Slide(ft)	57 (2.6%)	Azimuth	145.39 deg
300	ROP AVG	17.14 ft/hr		Hole Size	Total Drilled (ft)	2223 ft	TVD	5112.60
Time		Depth (ft)		Drilling Mode	Toolface	Slide Dist.	Rotate Drilled	ROP ft/hr
From	To	From	To				Drill Time (Hrs)	RPM
23:00	23:55	5148.00	5184.00	Rotary			36	39.3
0:17	4:20	5184.00	5247.00	Rotary			63	15.6
4:57	8:47	5247.00	5311.00	Rotary			64	16.7
9:34	12:34	5311.00	5374.00	Rotary			63	21.0
13:13	17:20	5374.00	5437.00	Rotary			63	15.3
19:01	21:30	5437.00	5483.60	Rotary			46.6	18.8
							2.48	80
								8-12
								600
								2150
								2250
								165
								112
								135
								8.2
								9.4
								8.3
								9.5
								8.3
								9.5
								8.4
								9.6
								8.5
								9.7

HALLIBURTON

Drilling Operation: Lesson Learn

Lesson Learnt Event

Jar Extended – PHR

- Several overpull was observed during POOH (up to 15 klbs)
- On surface observed jar piston extended which result unlatched piston.
- Re-Latching to normal position can be done on the rig floor by applying weight above jar 7-10klbs while jar clam on rotary table.





THANK YOU